

# Multi-Mode Microphone with I<sup>2</sup>S Digital Output

## GENERAL DESCRIPTION

The ICS-43434 is digital I<sup>2</sup>S output bottom port microphone. The complete ICS-43434 solution consists of a MEMS sensor, signal conditioning, an analog-to-digital converter, decimation and antialiasing filters, power management, and an industry standard 24-bit I<sup>2</sup>S interface. The I<sup>2</sup>S interface allows the ICS-43434 to connect directly to digital processors, such as DSPs and microcontrollers, without the need for an audio codec in the system.

The ICS-43434 has multiple modes of operation: High Performance, Low Power (AlwaysOn) and Sleep. The ICS-43434 has high SNR and 120 dB SPL AOP in all operational modes.

The ICS-43434 has a high SNR of 65 dBA and a wideband frequency response. The sensitivity tolerance of the ICS-43434 is  $\pm 1$  dB, which enables high-performance microphone arrays without the need for system calibration.

The ICS-43434 is available in a small 3.50 mm  $\times$  2.65 mm  $\times$  0.98 mm surface-mount package. The ICS-43434 is function-compatible with the ICS-43432 while providing equivalent electro-acoustic performance at lower power consumption and in a smaller package.

## APPLICATIONS

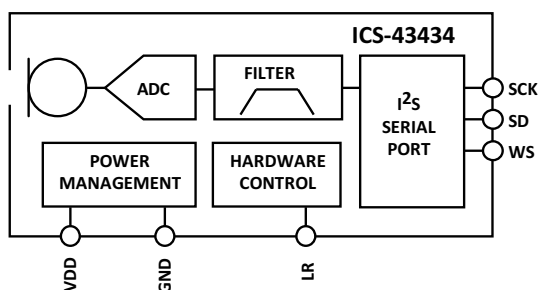
- Wearables
- Smart Televisions
- Remote Controls
- IoT Devices
- Teleconferencing Systems
- Gaming Consoles
- Security Systems

## FEATURES

| SPEC               | HIGH PERFORMANCE MODE | LOW-POWER MODE       |
|--------------------|-----------------------|----------------------|
| <b>Sensitivity</b> | -26 dB FS $\pm 1$ dB  | -26 dB FS $\pm 1$ dB |
| <b>SNR</b>         | 65 dBA                | 64 dBA               |
| <b>Current</b>     | 490 $\mu$ A           | 230 $\mu$ A          |
| <b>AOP</b>         | 120 dB SPL            | 120 dB SPL           |
| <b>Sample Rate</b> | 23 – 51.6 kHz         | 6.25 – 18.75 kHz     |

- Digital I<sup>2</sup>S interface with high precision 24-bit data
- Wide frequency response from 60 Hz to 20 kHz
- High power supply rejection: -100 dB FS
- Small 3.50 mm  $\times$  2.65 mm  $\times$  0.98 mm surface-mount package
- Compatible with Sn/Pb and Pb-free solder processes
- RoHS/WEEE compliant

## FUNCTIONAL BLOCK DIAGRAM



## ORDERING INFORMATION

| PART            | TEMP RANGE     | PACKAGING       |
|-----------------|----------------|-----------------|
| ICS-43434       | -40°C to +85°C | 13" Tape & Reel |
| EV_IC3-43434-FX | —              |                 |